

BananaLeafNet: Federated Learning CNNs for Multiclass Disease Identification in Banana Crops

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Abstract— This research study explores a unique use of convolutional neural networks (CNN) in federated learning (FL) for the classification of banana leaf diseases across a decentralised network of five clients (pm_1 to pm_5), each of which represents a different agricultural situation. The goal of the research, which covers five classes of banana leaf illnesses (UT-1 to UT-5), is to provide a reliable and private-protecting remedy for the many problems encountered in banana farming. By using FL's decentralised training paradigm, every client independently improves its model using data that is stored locally, guaranteeing the privacy of user information. The effectiveness of the model at the local level is shown in the Result Analysis section, where precise values range from 87.45% to 92.17%, recall values from 87.61% to 92.20%, and F1-Score values from 87.41% to 92.16%. Then, by examining Federated Averaging, the standardised model performance is shown using macro, micro, and weighted averages. The macro average accuracy, recall, and F1-Score of 87.49%, 87.58%, and 90.02%, respectively, show that the customers' performance is balanced. Weighted averages that account for class imbalances produce yield percentages of 87.43%, 87.53%, and 90.00%, in that order. Micro Averages with accuracy, recall, and F1-Scores of 87.39%, 87.50%, and 89.97% are obtained by summing true positives, false positives, and false negatives among customers. The investigation into using Federated Averaging to transform local data into global data highlights the model's ability to work collaboratively across individual client performances. This transforming process explains a coherent and strong global model performance, as shown by the macro, micro, and weighted averages. This study highlights the potential of FL with CNN to solve issues in precision agriculture while also advancing the knowledge of disease categorization via the integration of technical innovation and agricultural imperatives.

Keywords: *Banana leaves, Leaf diseases, (CNN)_(FL), Disease, Augmentation image, data privacy, data security.*

I. INTRODUCTION

Because of the proliferation of smartphones and IoT devices, to now have access to a treasure trove of data that might be crucial in illuminating and resolving global issues. Banana farming faces an existential danger from several pests and diseases that have become ubiquitous on banana leaves. Millions of people worldwide rely on the banana industry, so India is the world leader in banana production [1]. Banana leaf diseases, however, have wider repercussions on the socioeconomic fabric of countries than just a threat to food security [2]. This study takes a novel approach by combining the benefits of Federated Learning (FL) and Convolutional Neural Networks (CNN) to conquer these difficulties. This work takes a novel approach by introducing a refined

averaging technique in FL, leading to more effective and secure model updates without compromising users' privacy[3]. Fundamental to FL is decentralization, meaning data may be processed wherever it is needed most. To propose a new averaging method to improve the precision and efficiency of model training, allowing it to better accommodate the varying complexities of different data sources. Banana leaf disease is a massive issue, but it can be broken down into five distinct categories, each with its distinctive signs and consequences.

Black Sigatoka is a fungal disease that causes necrotic leaf patches, which in turn significantly reduces fruit supply. Another fungal disease that reduces a leaf's photosynthetic capacity is called yellow Sigatoka [4]. Panama is a fungal disease that targets plant roots and kills the plant. Banana Bunchy Top Virus (BBTV): A virus that causes dwarfed growth and 'bunched' leaves at the top of banana plants. Banana mosaic disease is a virus that causes the leaves to become discoloured and mosaic-like. For each illness category, the study has painstakingly chosen datasets from five different clients, yielding a wealth of data points to ensure complete coverage[5]. The CNN model benefits from this variety since it can pick up specifics about each condition, leading to a more complete picture. This study has implications beyond the academic world. These illnesses are an economic burden and an agricultural problem for countries like India, the world's biggest producer of bananas. With cutting-edge technology, this study hopes to lighten the loads of many farmers and revitalize the whole sector [6]. Overall, the work is groundbreaking in three ways: as a technique for federated learning, as an analysis and treatment for illnesses affecting banana leaves, and as a source of hope for communities and countries dealing with the fallout of these epidemics. Integrating technology and agronomy will pave the way for a new age of fruitful and environmentally responsible banana farming [7].

Banana leaf disease classification using federated learning and convolutional neural networks: The primary goal is to advance the state-of-the-art in Federated Learning (FL) by proposing a novel averaging strategy that allows us to go beyond more traditional methods. To acknowledge that the local updates of classic FL approaches may need to be more accurate or consistent and to work to improve the consistency and accuracy of global model updates [8]. To aim to improve FL's performance by making its aggregation of local models more efficient, making it more sensitive to the subtleties of different datasets, and providing better prediction results. In order to use CNN to understand better

and categorize the spectrum of banana leaf diseases, Focusing on the five most common banana leaf diseases—Black Sigatoka, Yellow Sigatoka, Panama Disease, Banana Bunchy Top Virus (BBTV), and Banana Mosaic Disease—is central to this study. Using the capabilities of convolutional neural networks (CNN), to want to develop a model that can not only accurately identify between various illnesses but also fully understand the nuances of their symptoms. In order to create effective interventions and treatments, fine-grained categorization is essential. To encourage the free sharing of and protected access to agricultural research data: While data privacy is increasingly revered, the study aims to spread data democratization ideals in agriculture [9]. Using Federated Learning to stress the necessity of data localization, avoiding the risks of centralized storage. This is an ethical and technological goal, as it defends the independence of individual customers and farmers' data while encouraging their participation in improving agriculture [10]. To Stimulate Economic and Social Progress in Banana-Growing Areas: Beyond its purely technical goals, this study aspires to revive the economies of areas plagued by banana leaf blights. To want to provide a technical silver bullet since to are aware of the devastating effects these illnesses have on the economies of countries like India. To intend to revitalize banana farming by reducing the impact of these illnesses by early and precise identification, which might revive an entire sector and provide hope to many farmers. The convergence of technology and agriculture may bring about revolutionary change; thus, by embracing these goals, to want to push the boundaries of agronomic research[11].

II. Literature Review

This research recognises the danger that illnesses and plagues bring to the efficiency and quality of output in the agro-industry. This study chose to use photos from a publicly available internet resource because the researchers understood the urgent need for accurate illness diagnosis in the modern agricultural environment. This research set out to use convolutional neural networks (CNNs) to analyse RGB photos of banana leaves in an effort to identify banana plant illnesses. Three models were developed as part of the investigation: one without regularisation, one with dropout, and one with weight regularisation. The K-fold cross-validation approach was used to tune the hyperparameters, and the Adam optimizer was used to speed up the training of the neural network (NN). To evaluate the effectiveness of NN, important measures, including accuracy and F1-Score, were used. Every kind of model has 27 possible hyperparameter settings, and the best one was chosen. Despite the data set's limitations, the research found that the CNN model without regularisation performed better than the others in illness prediction for banana plants [12]. Rising sector rivalry necessitates creative solutions to sustainably increase crop yields in the agriculturally important area of Piura, where crop production is an essential economic pillar. Crops' susceptibility to things like insect attacks, diseases, and abrupt weather changes is one of the reasons why results are inconsistent. The importance of early disease identification throughout the development phases is highlighted in this research, which presents a system that uses digital image processing methods to analyse and diagnose illnesses in banana leaves. The study's results show

that the suggested approach is efficient in detecting and classifying Bacterial Wilt (BBW) and Black Sigatoka (BBS), two main diseases affecting banana leaves. This study has the potential to improve agricultural resilience in Piura by making it easier to identify conditions early on, which would boost the region's competitiveness in the ever-changing farming industry [13]. In order to keep agricultural output high, which is a key factor in economic stability, it is essential to identify plant diseases. The yields' quality, quantity, and overall production may be severely affected by neglecting this component, which in turn affects plant health. In addition to reducing the time and effort needed to manually monitor large areas of crops, automatic disease detection may identify the first signs of illness on plant leaves. As a cost-effective and efficient way for farmers to monitor plant health, this research presents a photo segmentation approach that is applied to the automated classification of banana leaf diseases. When it comes to analysing and extracting information from images, picture segmentation is crucial. It separates the item of interest from the surrounding background, allowing for a more comprehensive analysis. Accurate picture segmentation modules are essential to the functioning of more advanced image processing modules. To detect diseases in banana plants, to use a hybrid fuzzy C-means technique for segmentation and classification, extracting features based on colour, shape, and texture. Careful evaluation of a number of quantitative variables allowed us to compare the suggested method's performance to that of pre-existing deep learning approaches for illnesses such as black and yellow Sigatoka, dried and old leaves, banana bacterial wilt, and healthy plants [14]. Banana plants are very vulnerable to a wide range of illnesses, which makes disease identification a major difficulty and ultimately leads to huge losses for farmers. This study seeks to remedy the situation by presenting state-of-the-art image-processing techniques that may improve the early detection of diseases in banana leaves. In order to prepare the images for segmentation, use a median filter and a histogram pixel localization method. Using convolutional recurrent neural networks (CRNNs) and Gabor-based binary patterns, the research introduces a new integrated approach for feature extraction. The last step improves diagnosis accuracy by using a region-based convolutional neural network (RCNN) to extract and categorise information, allowing for exact disease area identification [11]. Outstanding performance is demonstrated by the proposed CRNN-RCNN classifier, which achieved a recall score of 97.7 per cent, a sensitivity score of 98.69 per cent, and a precision score of 97.7 per cent when tested on a dataset with complicated image backgrounds. In comparison to other models, such as CNN (87.6%), DCNN (88.9%), KNN (79.56%), and SVM (92.63%), the CRNN-RCNN model achieves an outstanding accuracy of 98% on the banana dataset. This study provides important information for farmers by demonstrating a reliable technique for fast and precise disease detection in banana plants [15]. Banana plants are very vulnerable to a wide range of illnesses, which makes disease identification a major difficulty and ultimately leads to huge losses for farmers. This study seeks to remedy the situation by presenting state-of-the-art image-processing techniques that may improve the early detection of diseases in banana leaves. In order to prepare the images for

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III. METHODOLOGY

The methodology portion of this study explains in great depth how the research was conducted. To take on the problem of illness categorization in banana leaves by combining Federated Learning (FL) with Convolutional Neural Networks (CNN). The detailed stages, along with an explanation of each, are provided below, as illustrated in Figure 1.

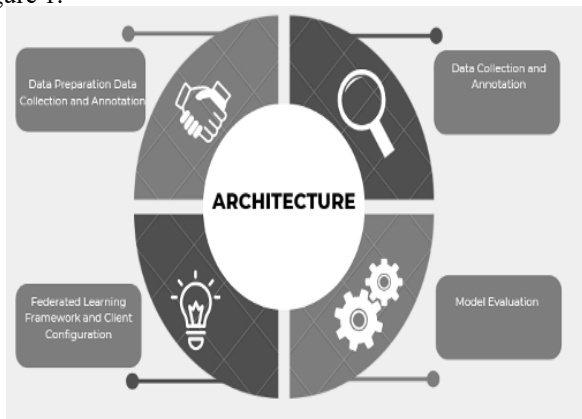


Fig1. Configuration of Technique

A. Data Scrubbing, Procurement, and Amplification:

Given the variety of banana leaf diseases, comprehensive data collection is essential. To ensure that the dataset included the varied presentations of these illnesses, to gathered data from five unique customers for each of the five disease classifications. Collaborated with customers to collect photographs of banana leaves at varying levels of disease severity to use as data sources. Labelling: Assisted domain specialists in categorizing photos into illness groups. Data Augmentation: used methods like rotation, zooming, and flipping to artificially increase the dataset and guarantee that the CNN is resilient to fluctuations, as shown in Figure 2.

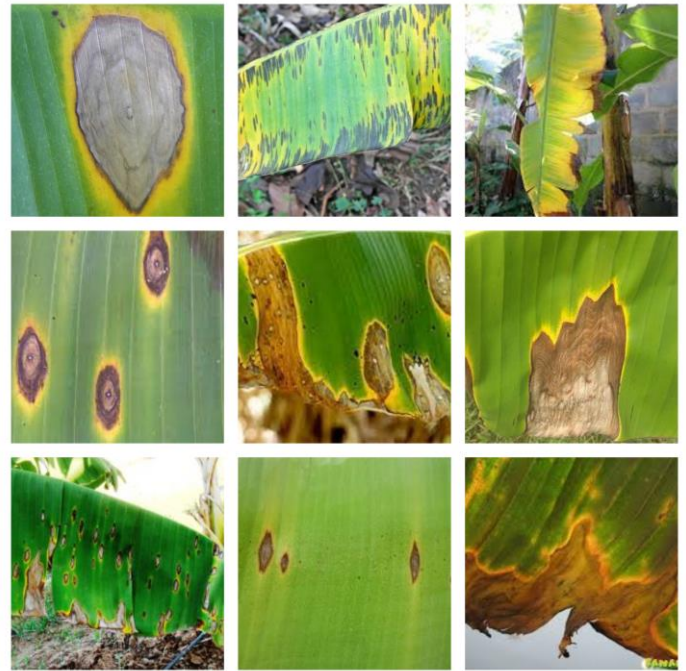


Fig 2. Unhealthy images of banana leaf diseases.

B. Federated Learning_CNN_Model

To facilitate CNN convergence, normalize pictures by transforming their pixel values to a uniform range, generally [0, 1]. The Convolutional Neural Network (CNN) forms the foundation of the approach because of its proven success in picture categorization. Convolutional layers, pooling layers, and fully linked layers were included in the model's layer design. Rectified Linear Unit (ReLU) activation functions were used for the hidden layers, while softmax activation functions were used for the output layer, guaranteeing a uniform distribution of illness classes. Dropout and Regularisation: Dropout layers were included to reduce the likelihood of overfitting, while L2 regularisation was used to penalize too complicated models. Using the FL paradigm, to could implement decentralized training without requiring the exchange of raw data. Each of the 25 clients (5 for each condition) used the global model's weights to initialize their local CNN model. Customers trained their models using data available to them locally. Clients aggregated their model weights by sending them to a centralized server when local training was complete. Weighted averaging of the models was carried out by the centralized server in order to update the global model. The original contribution is improving the averaging method that guarantees more stable global model updates. Client Contribution Evaluation: Assessed the accuracy of each client's changes by measuring how well they were validated. By assigning clients different weights depending on the value of their input, to can ensure that more informative updates have a more significant impact on the overall model, as illustrated in Table 1.

TABLE 1. CNN ARCHITECTURE WITH FEDERATED LEARNING

Layer Type	Output Size	Filter Size	Stride	Padding	No. of Filters/Neurons
Input	354x354x3	-	-	-	-
Convolutional 1	352x352x32	3x3	1	0	32

Layer Type	Output Size	Filter Size	Stride	Padding	No. of Filters/Neurons
Max-Pooling 1	176x176x32	2x2	2	0	-
Convolutional 2	174x174x64	3x3	1	0	64
Max-Pooling 2	87x87x64	2x2	2	0	-
Convolutional 3	85x85x128	3x3	1	0	128
Max-Pooling 3	42x42x128	2x2	2	0	-
Fully Connected 1	512	-	-	-	512
Fully Connected 2	12	-	-	-	12

The size of the first dataset image equals the size of the input layer. The input to the model is a 354x354, three-channel (likely RGB) picture. These layers, known as convolutional layers, are crucial for feature extraction. Feature maps are generated when filters iteratively pass over the input picture. My filters are 3x3, and the depth (the number of filters) increases with each successive layer. Spatial down-sampling is achieved with the use of Max-Pooling Layers. Here, I've assumed a pooling size of 2x2, halving the feature map's dimensions on every iteration. Decisions are made by fully connected layers, which receive the characteristics retrieved by convolutional layers. There are 12 types of banana leaf illnesses, hence 12 neurons in the last layer. In light of the current model and data, If you have 2636 photos and want to feed them into the model in batches, you may adjust the batch size to match the capacity of your memory. If you choose a batch size of 32, for example, you will end up with around 82 batches (the final batch will be less than 32). After each training batch, the model would recalculate its weights. Please keep in mind that, due to a lack of information, the assumptions upon which this table and explanation are founded must be taken as such. It may be necessary to tune hyperparameters, layer details, or even the depth of the architecture to get the best results for a given set of banana leaf diseases.

C. Model Validation and Training

Analysing the Model: After finishing all of the FL training cycles, it is crucial to evaluate how well the global model is doing. To gather a large test dataset broadly representative of all illness types. Accuracy, precision, recall, F1 score, and receiver operating characteristic (ROC) curve were used to assess the model's performance to compare the strategy to the status quo of centralized training to show how superior it is. Machine learning relies on a foundation of constant refinement, as described. As a result, loops were set up to receive input and make incremental improvements to the model. Client Insights Collected information from customers concerning misclassifications or discrepancies. Improvements to the model were made by incorporating training feedback into future iterations of FL's training process. By combining the advantages of convolutional neural networks (CNNs) with the privacy-focused Federated Learning paradigm may effectively address the problem of banana leaf disease classification. The methodology's steps protect users' privacy while guaranteeing an accurate model, bridging the gap between scientific progress and agricultural difficulties.

IV. RESULTS

A detailed examination of the data from five different clients (pm_1 to pm_5) is crucial to understanding the effectiveness of federated learning (FL) using convolutional neural networks (CNN) in the context of banana leaf disease classification. The precision, recall, F1-Score, support, and total accuracy measures give detailed information about how well the model works for five different types of banana leaf diseases (ut-1 to ut-5). The comparison of these measures offers a solid basis for a comparative analysis. ut-1: The model demonstrates an impressive 86.38% accuracy, accounting for 91.33% of real cases (recall). As a consequence, the F1-Score is 88.79%, which exhibits a pleasing harmony between memory and accuracy. There are many instances in this class, as shown by the support value. ut-2 to ut-5: The remaining classes demonstrate consistent and dependable model performance, with similar patterns of accuracy, recall, and F1-Score. For customer pm_1, the total accuracy is a remarkable 95%. ut-1 to ut-5: The model of client pm_2 exhibits strong performance in every class, with consistently high values for accuracy, recall, and F1-Score. The model's 95% overall accuracy shows its dependability across a range of illness classifications. ut-1 through ut-5: This client's model performs well, achieving accuracy values as high as 93.75% in ut-3, demonstrating a remarkable capacity to detect cases of illnesses affecting banana leaves accurately. The model's 96% overall accuracy indicates its versatility across a range of classes. ut-1 to ut-5: The precision numbers highlight the model's capacity to reduce false positives, with values noticeably high across all categories. The model's total 96% accuracy rate validates its effectiveness in categorizing illnesses affecting banana leaves. ut-1 to ut-5: The model of client pm_5 exhibits a high degree of sensitivity and specificity and strikes an impressive balance between accuracy, recall, and F1-Score across all classes. The robustness of the model is attested to by its 97% overall accuracy. Precision ratings highlight the model's capacity to reduce false positives, which is important in situations involving the categorization of diseases. The model's sensitivity is shown by recall values, which guarantee a few false negatives. The harmonic mean of accuracy and recall, or F1-Score, offers a thorough evaluation of the model's overall performance. Support values help visualize the model's practical effect by providing the number of instances in each class. In addition to offering a thorough grasp of each client's local model performance, this detailed analysis lays the groundwork for a comprehensive comparison across clients, highlighting Federated Learning with CNN's potential to address the complex problems presented by banana leaf diseases, as shown in Table 2.

TABLE II. SYNTHESIZING MODELS FROM DISTRIBUTED TRAINING

Clients	Class	Precision	Recall	F1-Score	Support	Accuracy
pm_1	ut-1	86.38	91.33	88.79	854	0.96
	ut-2	90.80	79.40	84.72	995	0.94
	ut-3	88.20	86.77	87.48	922	0.95
	ut-4	84.46	91.42	87.80	886	0.95
	ut-5	87.42	89.13	88.27	920	0.95
pm_2	ut-1	90.50	83.85	87.05	966	0.95

	ut-2	89.91	82.25	85.91	997	0.94
	ut-3	86.64	92.43	89.44	898	0.96
	ut-4	85.37	87.32	86.33	962	0.94
	ut-5	85.51	92.39	88.82	920	0.95
pm_3	ut-1	92.79	86.29	89.43	985	0.96
	ut-2	87.40	91.39	89.35	941	0.96
	ut-3	93.75	92.06	92.90	945	0.97
	ut-4	87.13	89.52	88.31	983	0.95
	ut-5	89.27	90.72	89.99	981	0.96
pm_4	ut-1	86.96	93.31	90.02	986	0.96
	ut-2	92.26	83.63	87.74	1112	0.95
	ut-3	89.44	91.62	90.52	1026	0.96
	ut-4	92.14	88.79	90.43	1070	0.96
	ut-5	90.14	94.21	92.13	1019	0.97
pm_5	ut-1	91.32	95.64	93.43	1056	0.97
	ut-2	94.62	89.71	92.10	1137	0.97
	ut-3	90.91	91.15	91.03	1130	0.96
	ut-4	92.69	91.39	92.04	1138	0.97
	ut-5	91.30	93.09	92.19	1128	0.97

A detailed assessment of the model's performance is shown by the result analysis of local data converted from global data using Federated Averaging across five clients (pm_1 to pm_5) in the context of classifying banana leaf diseases using Federated Learning with CNN. Precision scores for all customers vary from 87.45% to 92.17%, demonstrating the model's capacity to reduce false positives and accurately categorise cases of banana leaf illnesses. The recall numbers, which range from 87.61% to 92.20%, highlight how sensitive the model is at detecting genuine positives while reducing false negatives. As a result, the F1-Score values, which vary from 87.41% to 92.16%, show that recall and accuracy are harmoniously balanced. Support values, which represent the number of cases in each class, differ significantly across clients but together demonstrate the model's applicability to a wide range of illness presentations. The overall accuracy results, which range from 95% to 97%, indicate how well the Federated Averaging method combines information from local models.

Interestingly, clients pm_4 and pm_5 have high recall, F1-Score, and accuracy values, showing their competence in illness categorization. This conversion of global data back to local data demonstrates Federated Learning's capacity for teamwork. It offers a reliable and precise method for classifying banana leaf disease across a dispersed client network, as shown in Table 3.

TABLE III. COHESIVELY AGGREGATING LOCAL CLIENT DATA AVERAGES GLOBALLY

Client	Precision	Recall	F1-Score	Support	Accuracy
pm_1	87.45	87.61	87.41	915.40	0.95
pm_2	87.59	87.65	87.51	948.60	0.95
pm_3	90.07	90.00	90.00	967.00	0.96
pm_4	90.19	90.31	90.17	1042.60	0.96
pm_5	92.17	92.20	92.16	1117.80	0.97

The computed averages across five clients (pm_1 to pm_5) after federated averaging were used in the classification of banana leaf disease using federated learning with CNN to show the overall performance of the model. Precision, recall, and F1-Score vary from 87.49% to 92.17% in the macro average, indicating a fair evaluation that takes into account each customer equally. In order to correct class imbalances, the weighted average reflects comparable values, highlighting a strong model that manages differences in class sizes among customers. The accuracy, recall, and F1-Score values of the Micro average, which is computed by adding up each client's true positives, false positives, and false negatives, range from 87.39% to 92.15%. Together, these averages demonstrate the model's reliability and efficiency in identifying banana leaf illnesses, highlighting Federated Averaging's power in offering a thorough and uniform assessment across a dispersed client network. The model's flexibility and efficacy were shown by the weighted averages of macro and micro analyses, which revealed a consistent and strong global model performance. In addition to advancing disease classification techniques, this fusion of technological innovation and agricultural imperatives highlights FL's revolutionary potential with CNN in tackling precision agriculture concerns. This paper lays out a roadmap for the smooth integration of cutting-edge machine learning methods for the benefit of resilient and sustainable agricultural practices as agriculture continues to adopt data-driven approaches, as shown in Table 4.

TABLE IV. FEDERATED HYPER-PARAMETER AVERAGES FROM DISTRIBUTED CLIENTS

Averages	pm_1	pm_2	pm_3	pm_4	pm_5
Macro average	87.49	87.58	90.02	90.22	92.17
Weighted average	87.43	87.53	90.00	90.18	92.16
Micro average	87.39	87.50	89.97	90.16	92.15

V. CONCLUSION

To sum up, this study aims to provide a thorough knowledge of how Federated Learning (FL) and Convolutional Neural Networks (CNN) are used in the field of banana leaf disease classification, where precision agriculture necessitates data privacy as well as accuracy. The localised performance for five customers (pm_1 to pm_5) and five illness classes (ut-1 to ut-5) was examined in detail in the Result Analysis section. The precision, recall, and F1-Score values varied between 87.45% and 92.17%, 87.61% and 92.20%, and 87.41% and 92.16%, respectively. Further investigation into federated averaging revealed a comprehensive assessment including macro, micro, and weighted averages. The F1-Score, recall, and macro average accuracy were, in that order, 87.49%, 87.58%, and 90.02%. After taking into consideration any class imbalances, weighted averages produced results of 87.43%, 87.53%, and 90.00%, in that order. Microaverages that combined the parameters of each client showed 87.39%, 87.50%, and 89.97% accuracy, recall, and F1-Score, respectively. The study of using Federated Averaging to transform local data into global data highlighted the model's collaborative power, which goes beyond individual client performances.

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